

Cost, Freight and Delivery Analysis for Health Supply Chain

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Problem statement

Between 2006 and 2015, a USAID programme bought HIV and malaria medicines and test kits and shipped them to clinics in 43 countries. The programme spent about 1.7 billion US dollars. Every dollar spent on transport or lost to late delivery is a dollar that does not reach patients.

A supply chain manager running this programme would need clear answers to four business questions.

Cost. What it really costs to get medicine to a country, and how much of that is transport.

Shipment size and mode. Whether bigger shipments lower the cost per kilogram, and when it is worth paying for air instead of ocean.

Reliability. How often deliveries arrive on time, how long orders take, and which suppliers can be trusted.

Prices. Whether medicine prices fell over time, and how much more branded medicines cost than generics.

The goal of this project is to answer these questions with the data, show the limits of each answer, and turn the results into actions a manager could take.

The dataset

The data is the Supply Chain Shipment Pricing Data published by USAID under its Supply Chain Management System project. It is free to use under a Creative Commons licence.

It has 10,324 rows and 33 columns. Each row is one shipment line, which means one product sent to one country in one shipment. The columns cover the country, the supplier, the product and its group, the brand, the shipment mode, the order and delivery dates, the number of packs, the product value, the weight, the freight cost and the insurance cost.

Most of the money went to HIV medicines, which make up 8,550 of the lines and about 1.4 billion US dollars of product value. HIV test kits are the next largest group. Malaria medicines and malaria test kits make up only a small part of the data.

Challenges and assumptions

The raw data could not be used as it was. Each problem below needed a decision, and every decision is recorded in the Cleaning Log tab with the number of rows it affected.

Dates in two formats. Some dates were written only in numbers and others had the month as a word, and Excel read some of them wrongly. I imported the dates as text and rebuilt them with formulas so every date is correct.

Freight recorded on another line. On 2,445 lines the freight cell did not hold a number. Instead it pointed to another line in the same shipment that held one bill for all of them. I traced each line to that bill and split the bill across the lines.

How to split a shared bill. I split shared bills by the weight of each line, not by its value, because freight companies charge by weight. A value split would undercharge heavy, cheap items like oral liquids by more than half. To test this, I estimated the weight of each line and checked it against the real total shipment weight. The middle result was 1.01, which means the estimates add up almost exactly.

Freight hidden in the product price. On 1,442 lines the supplier included freight in the price of the medicine. Setting freight to zero would make these shipments look free to move, so I kept them in all totals and estimated their freight using the freight rate of other lines with the same shipment mode.

Freight billed elsewhere. On 344 lines the freight bill is not in the file at all. I left these out of freight rates instead of guessing.

Missing weight and insurance. Some lines had no weight, so freight per kg uses only lines with both a freight bill and a weight. 287 lines had no insurance value, so I estimated it using the typical insurance rate of the other lines, which is 0.16 percent of value.

Lead time is not available for every line. More than half the lines were sent from regional warehouses run by the programme and have no order date. Lead time is therefore measured only on lines sent directly from suppliers.

On time dates may not be reliable. 61 percent of lines arrived on exactly the scheduled date. That is unusually high, and it suggests some scheduled dates were changed after delivery. I report the on time rate both with and without these lines.

Possible duplicates. 12 rows looked like repeats, but each had its own ID. I flagged them instead of deleting them, since they may be real repeat orders.

Small groups. Some countries, product groups and shipment sizes have very few lines. Any group with fewer than 30 lines is marked in red, because its numbers can swing a lot.

After cleaning, I checked that the row count, total value, total packs and number of unique IDs all still match the raw file.

Calculations and why they matter

Landed cost

Landed cost is the product value plus freight plus insurance. It shows the true cost of getting medicine into a country, not just the price on the supplier invoice. A manager needs this number to set budgets and to compare suppliers fairly, because a cheap medicine from a far away supplier may cost more once it arrives.

Freight share of value

This is freight divided by product value, shown as a percent. It tells a manager how much of the budget goes to moving goods instead of buying them. Comparing it across shipment modes, countries and product groups shows where transport costs are highest and where savings are most likely.

Freight per kilogram by shipment size

This is total freight divided by total weight, grouped into five size bands from under 100 kg to over 10,000 kg. It answers a common planning question, which is whether it is cheaper to send one large order or several small ones. If the cost per kg drops sharply with size, combining orders saves money.

Air or ocean break even

Air costs more per kg but arrives sooner. Goods that arrive sooner tie up less money in stock, and that has a value. The model works out this saving as the value per kg of the goods, times a yearly holding cost of 20 percent, times the days saved, divided by 365. If the saving is bigger than the extra freight, air is worth it.

The break even value is the value per kg at which air and ocean cost the same. Goods worth more than this should fly; goods worth less should go by sea, unless there is an urgent need. The inputs can be changed on the tab, and the result updates by itself. A cost for a clinic running out of medicine can also be added.

On time rate and lead time

The on time rate is the share of lines delivered on or before the scheduled date. Lead time is the number of days from the purchase order to delivery. Together they tell a planner how early to place an order so clinics do not run out. I used the median lead time, the middle value, because a few very long orders would pull an average up.

Supplier scorecard

For the 20 suppliers with the most lines, the scorecard shows their number of lines, on time rate, average lead time and rank. A manager can use it to decide which suppliers to rely on and which ones to follow up with. A lookup lets you pick any supplier and see its value, first and last delivery and longest lead time.

Price per pack and price index

Price per pack is the total value divided by the total packs for one item in one year. To see how prices changed overall, I took the five long running HIV medicines and priced the same 2015 quantities at 2008 prices and at 2015 prices. Keeping the quantities fixed means the change comes only from prices, not from buying more or less. This works the same way as a consumer price index.

Branded versus generic

This compares the price of one tablet, capsule or ml for the same medicine, dose and form, sold as a brand and as a generic. Comparing like with like shows the real extra cost of a brand. I also compared only purchases made in the same year, so that brands bought in early years, when all prices were higher, do not make the gap look bigger than it is.

Findings

1. Freight adds a small share to cost, but it depends on the mode. Overall, freight added 4.5 percent to the value of the goods. Air added 7.3 percent, while truck added only 2.2 percent. Insurance added another 0.16 percent. Total landed cost was about 1.70 billion US dollars on 1.63 billion of product value.

2. Bigger shipments are much cheaper per kilogram. Air shipments under 100 kg cost 36.05 US dollars per kg. Air shipments over 10,000 kg cost 1.62 US dollars per kg. Ocean and truck follow the same pattern. Combining small orders where possible would lower freight costs.

3. Ocean is the better choice for most generic medicines. For shipments of 2,000 to 9,999 kg, air cost 3.53 US dollars more per kg than ocean. Air only pays for itself on goods worth more than about 88 US dollars per kg. Generic medicines in this size range were worth about 70 US dollars per kg, so they should go by sea unless they are urgent. Branded medicines at about 91 US dollars per kg and HIV test kits at about 96 sit just above the line.

4. The reported on time rate is likely too high. On paper, 88.5 percent of lines arrived on time. Without the lines that arrived on exactly the scheduled date, the rate is 70.4 percent. The real figure is likely somewhere between 70 and 89 percent. Lines from regional warehouses were on time 82.8 percent of the time, compared with 94.7 percent for lines sent directly from suppliers. The median lead time for direct orders was 92 days, so orders need to be placed about three months ahead.

5. Prices fell, but brands still cost much more. The five most shipped HIV medicines make up 44 percent of HIV medicine spending. They became 42 percent cheaper per pack between 2008 and 2015. Efavirenz alone fell by 74 percent. For the same medicine, dose and form, branded versions cost about 2.8 times the generic price, and the gap stays about the same when only same year purchases are compared. Choosing generics where they are approved is the clearest way to save money.

The dashboard

The dashboard puts the main numbers on one page for a manager who does not want to read the detailed tabs. Three dropdown filters on the left let you choose a year, a shipment mode and a product group. Every number and chart updates when a filter changes, and each filter can be set to All.

Number shown	What it tells you	Value with all filters set to All
Lines	How many shipment lines are in the current view	10,324
Product value	What the goods cost	1.63 billion US dollars
Landed cost	The full cost once freight and insurance are added	1.70 billion US dollars
Freight share of value	How much freight adds to the cost of the goods	4.5 percent
Freight per kg	The average transport cost for each kilogram	3.24 US dollars
On time rate	The share of lines delivered on or before the scheduled date	88.5 percent
Median lead time	Typical days from order to delivery for direct orders	92 days

Below the numbers are four charts. The first shows product value by year, so you can see how the programme grew. The second shows the freight share for each shipment mode, which makes the cost of air easy to see. The third shows the on time rate by year, with a clear drop in 2010 and 2011. The fourth shows landed cost by product group, which shows that HIV medicines take most of the budget.

Limits

The data has no ship date, so travel time cannot be separated from the time suppliers took to make and pack the goods. The air and ocean model counts only holding cost, not the full cost of a clinic running out of medicine. Lead time covers only lines sent directly from suppliers. The prices are those agreed for a large donor programme, not shop prices, and the price trend covers only medicines bought every year since 2008.